

U-Pb Geochronology: Dual Decay Schemes, The Wetherill Concordia Curve & Discordia Intercept Dynamics

Comprehensive Geochronology Guide · Authoritative References: K. C. Misra (2012, Ch. 10) & William M. White (2005, Ch. 8)

THEORETICAL FOUNDATION: THE COUPLED U-Pb CHRONOMETERS

Dual Radiogenic Decay, The Concordia Equation, Episodic Lead Loss & Discordia Intercept Interpretation

1. The Dual Uranium-Lead Clocks in a Single Mineral

The Uranium-Lead (U-Pb) isotopic system is the undisputed "gold standard" of high-precision geochronology. Its unique power stems from the fact that nature provides **two separate isotopes of the same parent element (Uranium) that decay to two separate isotopes of the same daughter element (Lead)** with drastically different half-lives:

THE TWO INDEPENDENT U-Pb DECAY EQUATIONS

Clock 1: ${}^{238}\text{U} \rightarrow {}^{206}\text{Pb} + 8\alpha + 6\beta^- + Q$

- Decay constant: $\lambda_{238} = 1.55125 \times 10^{-10} \text{ yr}^{-1}$ ($t_{1/2} = 4.468 \text{ Ga}$)

- Ratio equation: $\left(\frac{{}^{206}\text{Pb}^*}{{}^{238}\text{U}} \right) = e^{\lambda_{238} t} - 1$

Clock 2: ${}^{235}\text{U} \rightarrow {}^{207}\text{Pb} + 7\alpha + 4\beta^- + Q$

- Decay constant: $\lambda_{235} = 9.8485 \times 10^{-10} \text{ yr}^{-1}$ ($t_{1/2} = 0.704 \text{ Ga}$)

- Ratio equation: $\left(\frac{{}^{207}\text{Pb}^*}{{}^{235}\text{U}} \right) = e^{\lambda_{235} t} - 1$

Where ${}^{206}\text{Pb}^*$ and ${}^{207}\text{Pb}^*$ represent radiogenic lead atoms accumulated purely from in-situ decay since crystallization.

Because both decay reactions occur simultaneously inside the exact same mineral crystal lattice (predominantly **zircon, ZrSiO_4** , which incorporates abundant U^{4+} upon crystallization while strictly excluding initial Pb^{2+}), any closed-system zircon must record **identical ages** for both systems:

The Condition of Concordance:

A sample is **concordant** if and only if: $t_{206/\text{U}} = t_{207/\text{U}} = t_{\text{crystallization}}$.

2. The Wetherill Concordia Curve: Definition & Mathematical Locus

In 1956, George W. Wetherill introduced a brilliant graphical method to evaluate the concordance of U-Pb systems. Plotting the two radiogenic ratios against each other:

- **Y-axis:** $^{206}\text{Pb}^* / ^{238}\text{U} = e^{\lambda_{238} t} - 1$
- **X-axis:** $^{207}\text{Pb}^* / ^{235}\text{U} = e^{\lambda_{235} t} - 1$

PARAMETRIC DEFINITION OF THE CONCORDIA CURVE

The **Concordia Curve** is the parametric curve defined by:

$$x(t) = e^{\lambda_{235} t} - 1$$

$$y(t) = e^{\lambda_{238} t} - 1$$

For every single time t ranging from 0 to 4.55 Ga , there is exactly one point $(x(t), y(t))$ on this plane. This curve represents the **geometric locus of all points having identical $^{206}\text{Pb}/^{238}\text{U}$ and $^{207}\text{Pb}/^{235}\text{U}$ ages.**

*"Wetherill (1956) devised a graphical method that has proved very useful in evaluating U-Pb data. The curve on this plot is called the **concordia**, and is the locus of all points for which the $^{206}\text{Pb}/^{238}\text{U}$ age equals the $^{207}\text{Pb}/^{235}\text{U}$ age. Any sample that has remained closed to U and Pb since crystallization will plot on the concordia curve at a position corresponding to its age."*

— Kula C. Misra, *Introduction to Geochemistry: Principles and Applications* (2012), Chapter 10, p. 240.

3. Textbook Evidence: Concordia, Discordia & Lead Loss

Figure 10.10 from Misra illustrates the geometry of the Wetherill Concordia curve and the linear Discordia chord generated by episodic lead loss:

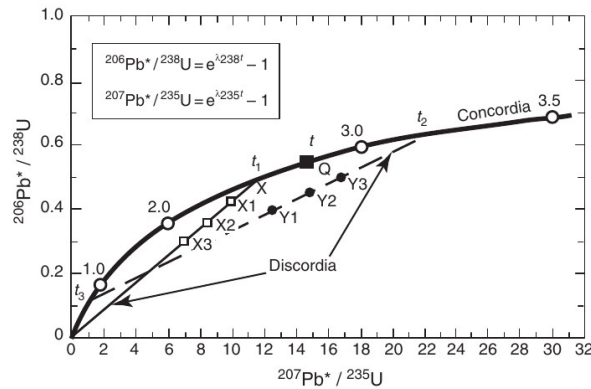


Fig. 10.7 U-Pb concordia diagram illustrating interpretations of lead isotopic data for uranium-bearing minerals such as zircon. The concordia is generated by plotting $^{206}\text{Pb}^*/^{238}\text{U}$ against $^{207}\text{Pb}^*/^{235}\text{U}$, which are calculated for selected values of t (a few of which are marked on the concordia), using equations (10.22) and (10.24). For a zircon plotting on the concordia (e.g.,

the crystallization age of zircon (t_2). Assuming the crystallization was instantaneous, the lower intercept is interpreted as the age (t_3) of the event that disturbed the system. If the discordia represents continuous diffusion over a long period of time, the intercept has no age significance if the Pb loss is episodic and continuous Pb loss models for discordias. The computer code *Isoplot* (version 2003) can be used for constructing the concordia and fitting of discordias to analytical data, and for determining the coordinates of the points of intersection with the concordia curve.

Isotopic composition of lead in U- and Th-bearing minerals and whole rocks

An equation relating the isotopic composition of lead in U- and Th-bearing minerals to model age can be derived by combining equations (10.21) and (10.22) and (10.23) and the present-day $^{238}\text{U}/^{235}\text{U}$ ratio, which is equal to 137.88.

Figure 1 (Misra Fig. 10.10, p. 240): The Wetherill Concordia-Discordia diagram. The curved solid line is the **Concordia Curve**—the locus of points with concordant ages ($t_{206} = t_{207}$). Zircons that experienced episodic lead loss at time t_2 plot below concordia along a straight chord called the **Discordia line**. The **Upper Intercept** with concordia reveals the true primary crystallization age (t_1), while the **Lower Intercept** dates the metamorphic/thermal lead-loss event (t_2).

4. Discordia Chords & Physical Causes of Discordance

When an igneous rock cools, its zircons start at time t_1 at a specific point on the Concordia curve. Over geologic time, if the zircons remain an undisturbed closed system, they move smoothly up along Concordia. However, zircons frequently suffer open-system disturbance:

Mechanisms of Lead Loss in Zircons:

- 1. Metamictization:** Alpha particles emitted by ^{238}U and ^{232}Th recoil through the zircon lattice, breaking chemical bonds and converting the crystalline structure into an amorphous, glass-like metamict state.
- 2. Episodic Thermal Resetting:** During a later metamorphic or tectonic heating event at time t_2 , radiant heat allows incompatible radiogenic lead ($^{206}\text{Pb}^{2+}$), which is an outcast in the $^{207}\text{Pb}^{4+}$ site) to diffuse out of damaged zones into surrounding fluids.
- 3. Continuous Diffusion vs. Recent Weathering:** Lead loss occurring during modern groundwater exposure causes points to align along a discordia chord that points directly toward the origin ($t_2 = 0$).

*"Because lead is lost in varying proportions from different zircon grains or zones, a suite of cogenetic discordant zircons defines a linear array—the **discordia chord**. The upper intercept with concordia dates the original crystallization of the magma, while the lower intercept dates the episodic lead loss event."*

— William M. White, *Geochemistry* (2005), Chapter 8, p. 320.

5. Comparison: Concordia Curve vs. Isochrons vs. Mixing Lines

Candidates must clearly differentiate between the four distinct linear and curvilinear constructs in isotope geochemistry:

Construct	Mathematical Shape	Plotted Coordinates	Physical / Geological Meaning
Concordia Curve (The Answer)	Exponential Curve	$\frac{{}^{206}\text{Pb}}{{}^{238}\text{U}}$ vs. $\frac{{}^{207}\text{Pb}}{{}^{235}\text{U}}$	Locus of points having identical $\frac{{}^{206}\text{Pb}}{{}^{238}\text{U}}$ and $\frac{{}^{207}\text{Pb}}{{}^{235}\text{U}}$ ages ($t_{206} = t_{207}$).
Discordia Curve / Chord	Straight Line (Chord)	$\frac{{}^{206}\text{Pb}}{{}^{238}\text{U}}$ vs. $\frac{{}^{207}\text{Pb}}{{}^{235}\text{U}}$	Linear array formed by discordant zircons that suffered open-system lead loss.
Isochron Line	Straight Line	Radiogenic isotope ratio vs. parent-daughter ratio (e.g., $\frac{{}^{87}\text{Sr}}{{}^{86}\text{Sr}}$ vs $\frac{{}^{87}\text{Rb}}{{}^{86}\text{Sr}}$)	Locus of cogenetic coeval samples having the same initial ratio and identical crystallization age.
Mixing Line	Straight Line or Hyperbola	Isotopic ratio vs. concentration or inverse concentration (δ/C)	Binary physical mixture of two distinct isotopic reservoirs (e.g., mantle melt + crust).



EXAM QUESTION & RIGOROUS OPTION-BY-OPTION DISSECTION

DGMS 2019 Examination · Official Question Statement, Verified Answer & Detailed Distracter Analysis

DGMS 2019 · QUESTION 23

In a $^{238}\text{U}/^{206}\text{Pb}$ versus $^{235}\text{U}/^{207}\text{Pb}$ ratio diagram, the locus of points having the same $^{238}\text{U}/^{206}\text{Pb}$ and $^{235}\text{U}/^{207}\text{Pb}$ ages is known as:

- (a) Mixing line
- (b) Isochron line
- (c) Concordia curve
- (d) Discordia curve

Correct Option: (c) Concordia curve



Verified via K. C. Misra (Introduction to Geochemistry, p. 240) & William M. White (Geochemistry, pp. 317–321)

INCORRECT Option (a): Mixing line

Detailed Refutation: A *mixing line* is a linear trend formed on a geochemical variation or isotope ratio plot when two distinct geochemical components or reservoirs (such as depleted mantle magma and upper crustal assimilate) are mechanically or magmatically blended together. It has no mathematical relation to the locus of concordant radio-decay ages.

INCORRECT Option (b): Isochron line

Detailed Refutation: An *isochron line* is a straight line on an isochron diagram (such as $^{87}\text{Sr}/^{86}\text{Sr}$ vs. $^{87}\text{Rb}/^{86}\text{Sr}$ or $^{143}\text{Nd}/^{144}\text{Nd}$ vs. $^{147}\text{Sm}/^{144}\text{Nd}$) formed by cogenetic rocks or coexisting minerals that crystallized at the same time from an isotopically homogeneous source. In contrast, the locus of points where two independent radioactive decay schemes yield identical ages on a Wetherill diagram is an exponential curve termed Concordia.

CORRECT ANSWER Option (c): Concordia curve

Detailed Validation: The **Concordia curve** (introduced by George Wetherill in 1956) is explicitly and uniquely defined in geochronology as the **parametric curve representing the locus of all points for which the ^{238}U to ^{206}Pb age equals the ^{235}U to ^{207}Pb age** ($t_{206} = t_{207}$). Any mineral grain (such as zircon, monazite, or baddeleyite) that has remained completely closed to loss or gain of uranium and lead since crystallization will plot directly on this curve at the point corresponding to its true crystallization age. This matches the exam question verbatim.

INCORRECT Option (d): Discordia curve

Detailed Refutation: A *Discordia* (or Discordia line / chord) is a straight chord connecting points that have suffered **open-system disturbance (such as lead loss or uranium gain)** and consequently have *discrepant, discordant ages* ($t_{206} \neq t_{207}$). The Discordia chord intersects the Concordia curve at an upper intercept (primary crystallization age) and a lower intercept (episodic disturbance age), but it is NOT the locus of identical ages.



SYNTHESIS & HIGH-YIELD PRACTICE QUESTIONS

Curated Problems Covering Tera-Wasserburg Diagrams, Intercept Mathematics & Metamictization

High-Yield Exam Practice Questions

Practice Q1 (Intercept Interpretation): Upper vs. Lower Intercepts

A suite of discordant zircons from an Archean gneiss defines a Discordia line that intersects the Wetherill Concordia curve at 2.70 Ga (upper intercept) and 0.60 Ga (lower intercept). What is the geological interpretation of these two dates?

Solution & Rationale:

- **Upper Intercept (2.70 Ga):** The true primary magmatic crystallization age of the protolith granite.
- **Lower Intercept (0.60 Ga):** The age of a high-temperature metamorphic or hydrothermal tectonic event (e.g., Pan-African orogeny) that triggered episodic open-system lead loss from the damaged zircon crystals.

Practice Q2 (Modern Weathering): Zero Lower Intercept

What does it indicate when a Discordia line passes directly through the origin (0 Ma) at its lower intercept?

Answer: It indicates that lead loss occurred **recently (continuous modern weathering or groundwater leaching)** during the present exposure of the rock at the surface, rather than during an ancient episodic metamorphic event.

Practice Q3 (Mineral Crystal Chemistry): Why Zircon is Ideal

Why is Zircon (ZrSiO_4) the premier mineral for U-Pb dating, whereas Galena (PbS) cannot be dated by Concordia?

Answer: Zircon readily accepts uranium (U^{4+} , ionic radius $1.00 \text{ \AA}$) in place of zirconium (Zr^{4+} , radius $0.84 \text{ \AA}$), but strictly rejects lead (Pb^{2+} , radius $1.29 \text{ \AA}$) due to its large size and lower charge. Thus, zircon has virtually zero initial common lead ($[\text{Pb}]_{\text{initial}} \approx 0$). In contrast, Galena accepts massive amounts of initial common lead, making it impossible to measure radiogenic accumulation.

Practice Q4 (Tera-Wasserburg Concordia): Inverse Coordinates

In modern in-situ microbeam dating (SHRIMP / LA-ICP-MS), what coordinate system is used on a Tera-Wasserburg Concordia diagram?

Answer: The Tera-Wasserburg diagram plots $\frac{{}^{207}\text{Pb}^*}{{}^{206}\text{Pb}^*}$ on the y-axis versus $\frac{{}^{238}\text{U}}{{}^{206}\text{Pb}^*}$ on the x-axis. It is especially advantageous because it does not require prior common-lead correction and is directly applicable to young zircons.

Practice Q5 (Metamict State): Radiation Damage

Describe the "metamict state" in radioactive minerals and explain how it facilitates discordance.

Answer: Alpha decay of uranium and thorium ejects high-energy alpha particles and recoiling heavy daughter nuclei, breaking structural Si-O and Zr-O bonds and converting the ordered crystalline lattice into an amorphous, porous, metamict glass. This creates microscopic diffusion pathways that allow radiogenic Pb^{2+} to escape, generating discordant ages.

Practice Q6 (Inherited Zircon Cores): Complex Polycyclic Ages

When an S-type crustal granite magma assimilates older metasediments, what U-Pb feature is commonly observed inside individual zircon crystals?

Answer: Inherited (detrital) zircon cores surrounded by younger magmatic overgrowth rims. The core yields an older concordant/discordant age corresponding to the sedimentary source terrane, while the rim dates the magmatic intrusion.

Practice Q7 (Decay Constants): Ratio of Half-Lives

Why does ${}^{235}\text{U}$ decay more than six times faster than ${}^{238}\text{U}$?

Answer: ${}^{235}\text{U}$ has an odd number of nucleons ($92\text{ protons} + 143\text{ neutrons}$), giving it lower nuclear binding energy than even-even ${}^{238}\text{U}$ ($92\text{ protons} + 146\text{ neutrons}$). Consequently, its decay constant is larger ($\lambda_{235} = 9.85 \times 10^{-10} \text{ yr}^{-1}$ vs. $\lambda_{238} = 1.55 \times 10^{-10} \text{ yr}^{-1}$), and its half-life is shorter (0.704 Ga vs. 4.468 Ga).

Practice Q8 (Uranium Gain vs. Lead Loss): Position Relative to Concordia

If a zircon grain experiences modern uranium gain instead of lead loss, where does its data point plot relative to the Concordia curve?

Answer: Below the Concordia curve (plotting in the same lower quadrant as lead loss), because adding denominator uranium (U) lowers both the $\frac{{}^{206}\text{Pb}^*}{{}^{238}\text{U}}$ and $\frac{{}^{207}\text{Pb}^*}{{}^{235}\text{U}}$ ratios simultaneously. Conversely, uranium loss or lead gain plots *above* the Concordia curve (reverse discordance).

Comprehensive U-Pb Geochronology Summary Matrix

Geochronological Feature	Mathematical Formulation	Physical / Geological Entity	Diagnostic Significance
Concordia Curve (The Answer)	$y(t) = e^{\lambda_{238}t} - 1$, $x(t) = e^{\lambda_{235}t} - 1$	Locus of identical dual ages	Defines undisturbed, closed-system magmatic crystallization ages.
Discordia Line	Linear chord connecting (x_1, y_1) and (x_2, y_2)	Straight chord through disturbed points	Upper intercept = crystallization age (t_1); Lower intercept = lead-loss event (t_2).
Primary Dating Mineral	ZrSiO_4 (Zircon)	Zircon, Baddeleyite, Monazite	Extremely high closure temperature ($>900^\circ\text{C}$); rejects initial lead.
Pb Loss Mechanism	Metamictization + Diffusion	Pb^{2+} expulsion from damaged lattice	Shifts data points downwards away from Concordia along Discordia chords.